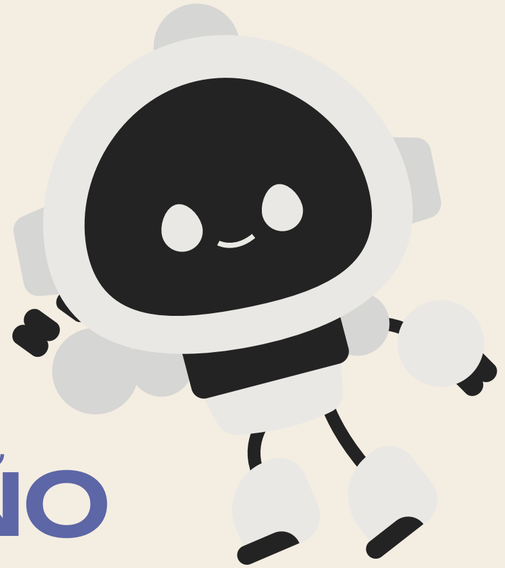
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IMPLEMENTACIÓN Y VERIFICACIÓN PRELIMINAR

GRUPO 1

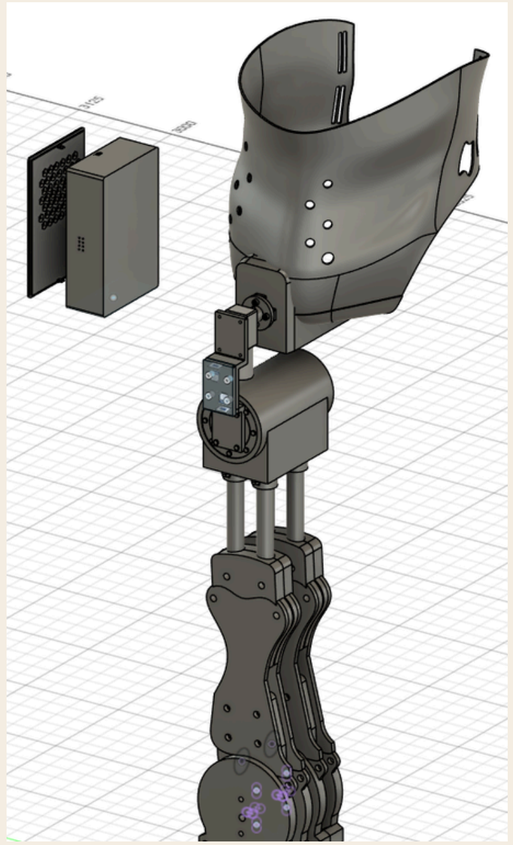


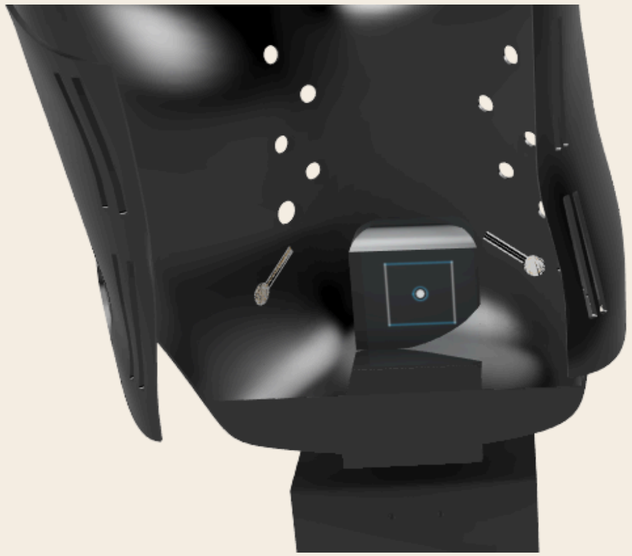
FUNDAMENTOS DE BIODISEÑO

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MODELO 3D COMPLETO





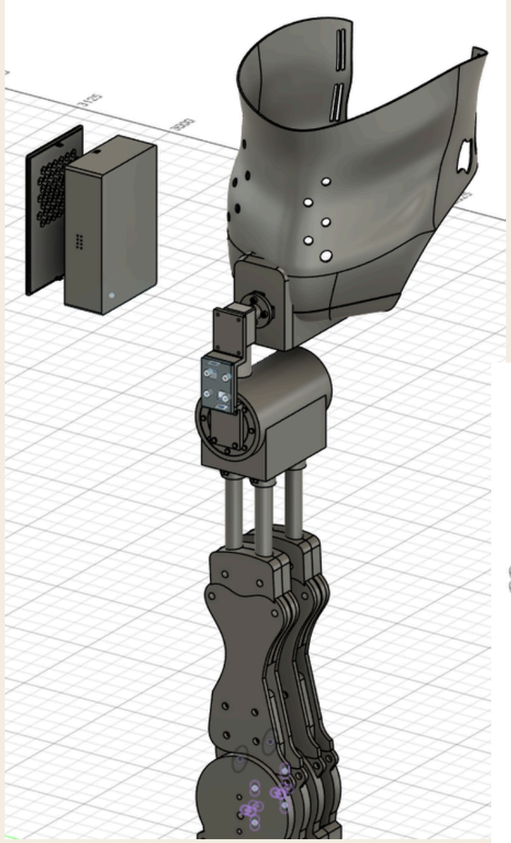


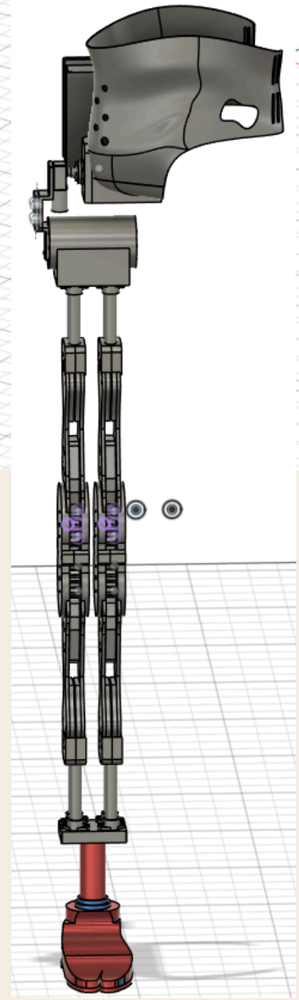


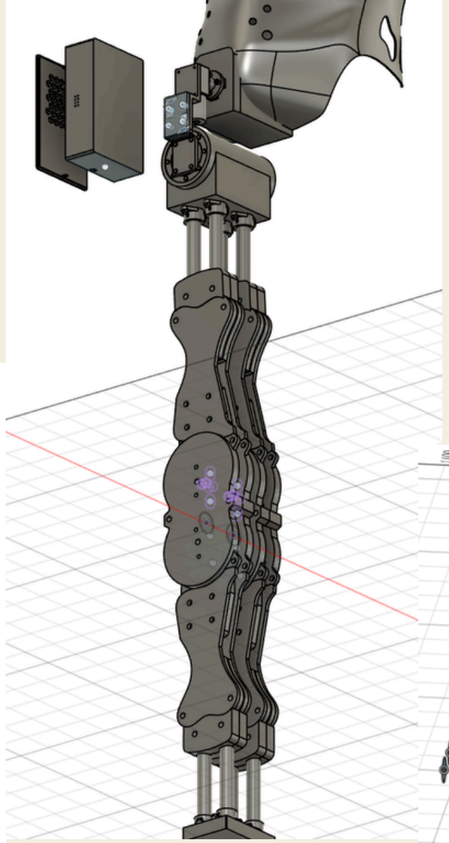
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HTTPS://A360.CO/4A2ETX6

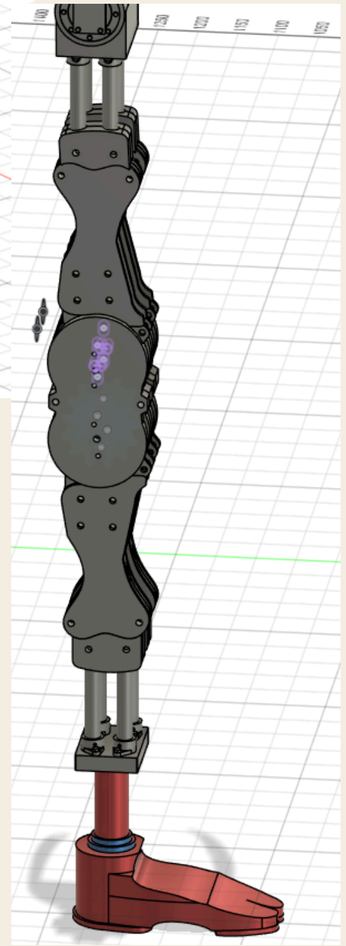
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MODELO 3D







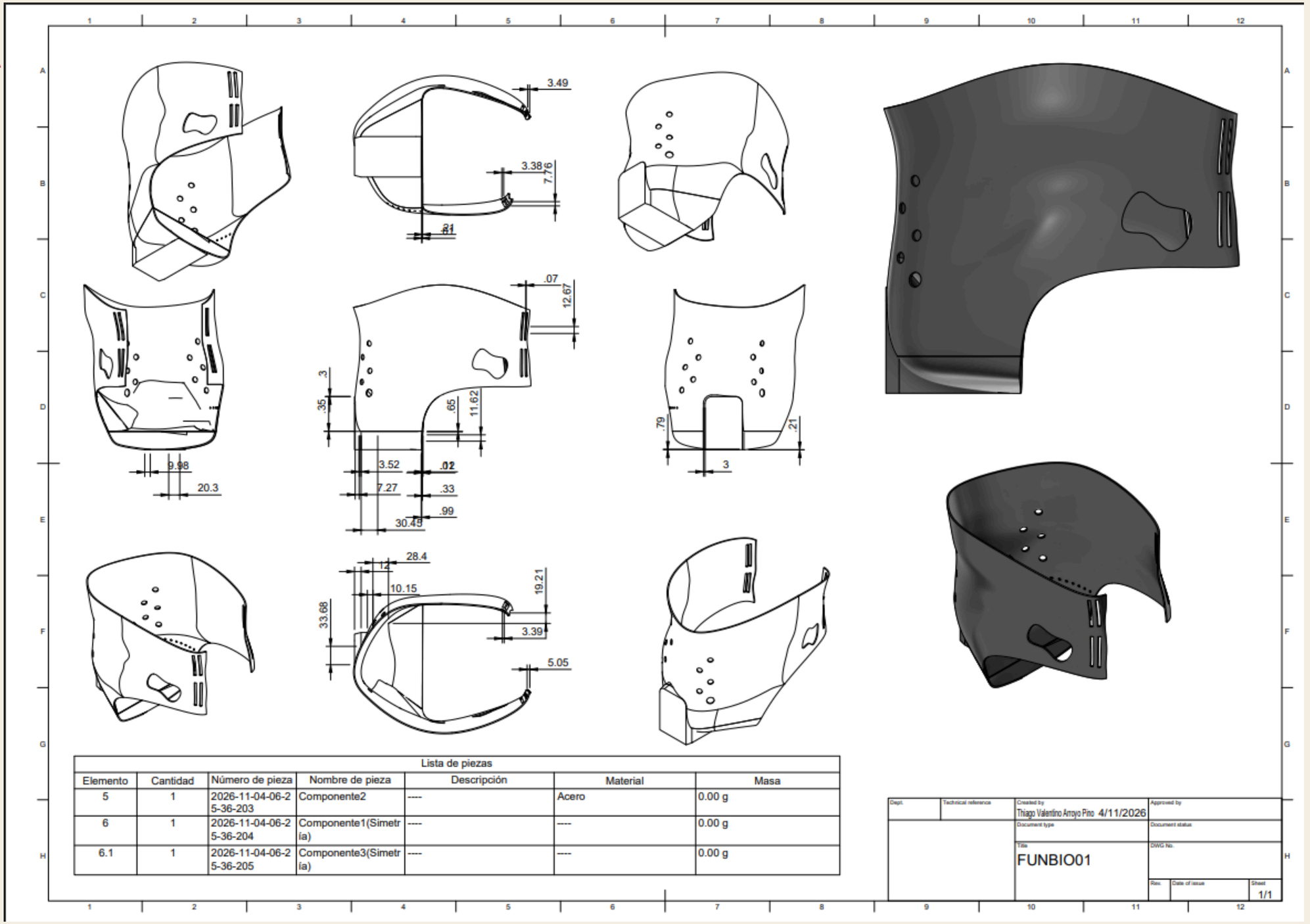


LINK:
HTTPS://A360.CO/4A2ETX6

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Socket:

PLANOS



Lista de piezas

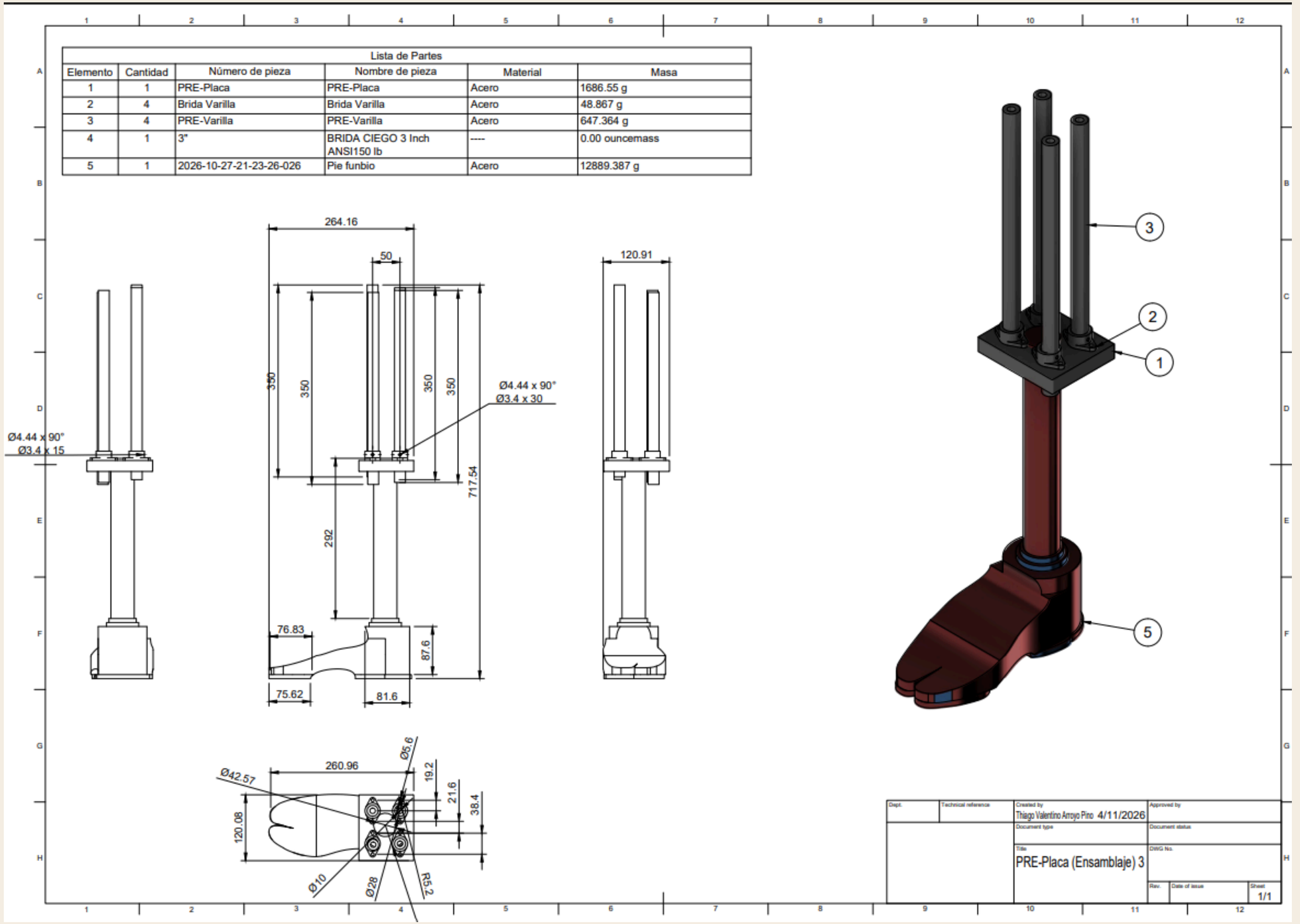
Elemento	Cantidad	Número de pieza	Nombre de pieza	Descripción	Material	Masa
5	1	2026-11-04-06-25-36-203	Componente2	----	Acero	0.00 g
6	1	2026-11-04-06-25-36-204	Componente1(Simetría)	----	----	0.00 g
6.1	1	2026-11-04-06-25-36-205	Componente3(Simetría)	----	----	0.00 g

Dept.	Technical reference	Created by: Thiago Valentino Arroyo Pino 4/11/2026	Approved by
		Document type	Document status
		Title: FUNBIO01	DWG No.
		Rev.	Date of issue / Sheet: 1/1

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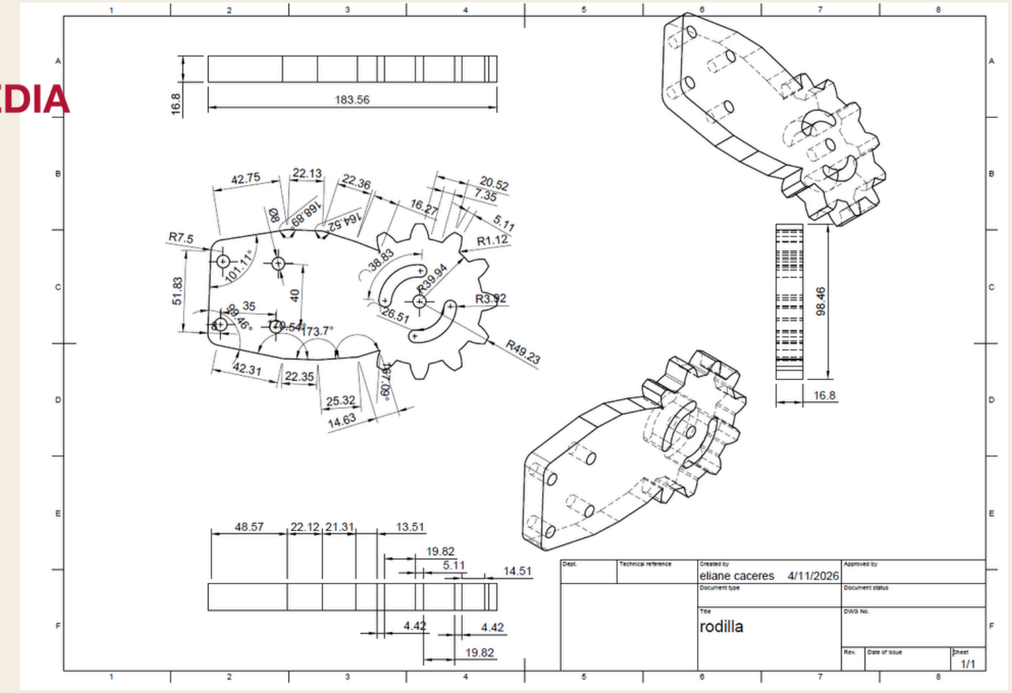
Pie:

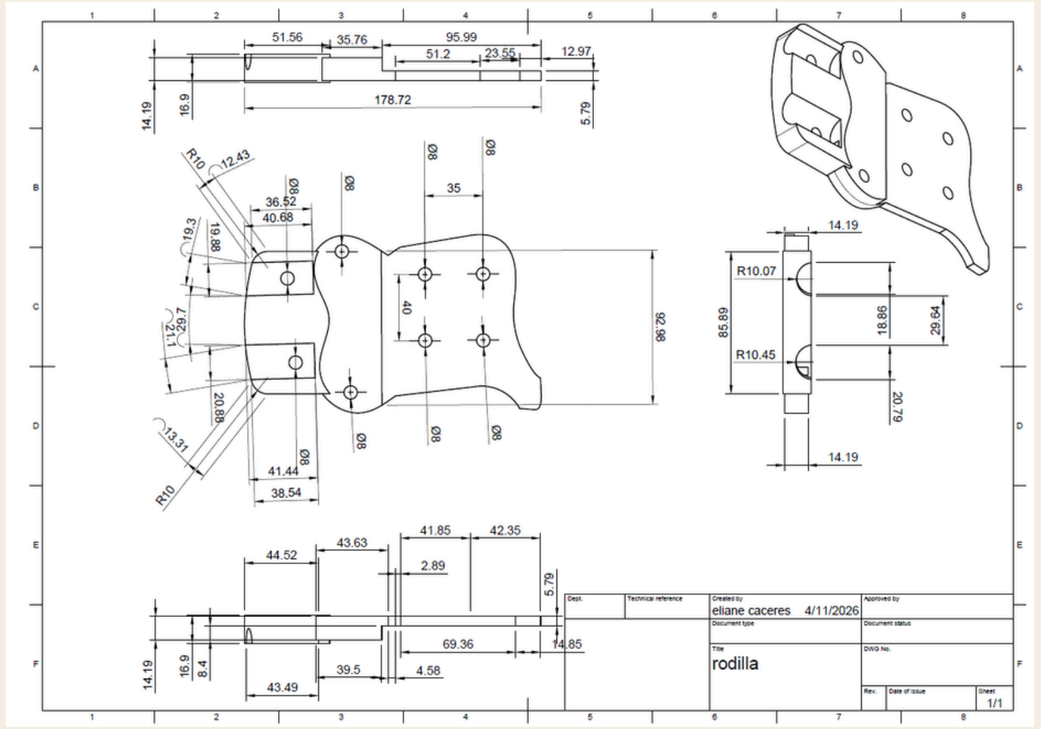
Lista de Partes					
Elemento	Cantidad	Número de pieza	Nombre de pieza	Material	Masa
1	1	PRE-Placa	PRE-Placa	Acero	1686.55 g
2	4	Brida Varilla	Brida Varilla	Acero	48.867 g
3	4	PRE-Varilla	PRE-Varilla	Acero	647.364 g
4	1	3"	BRIDA CIEGO 3 Inch ANSI150 lb	----	0.00 ouncemass
5	1	2026-10-27-21-23-26-026	Pie funbio	Acero	12889.387 g

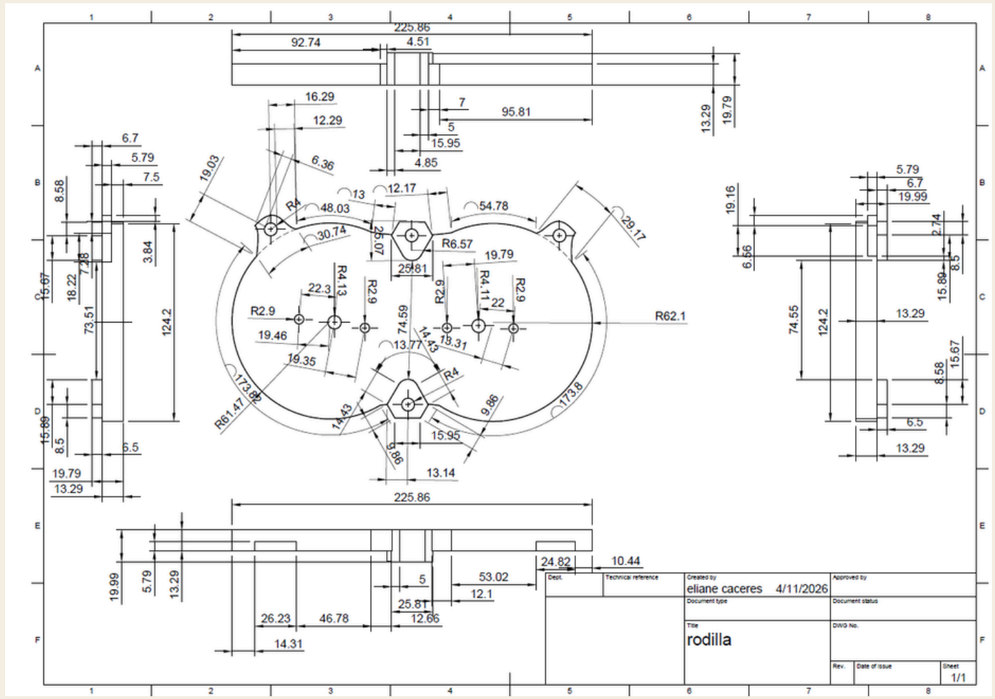


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Rodilla:

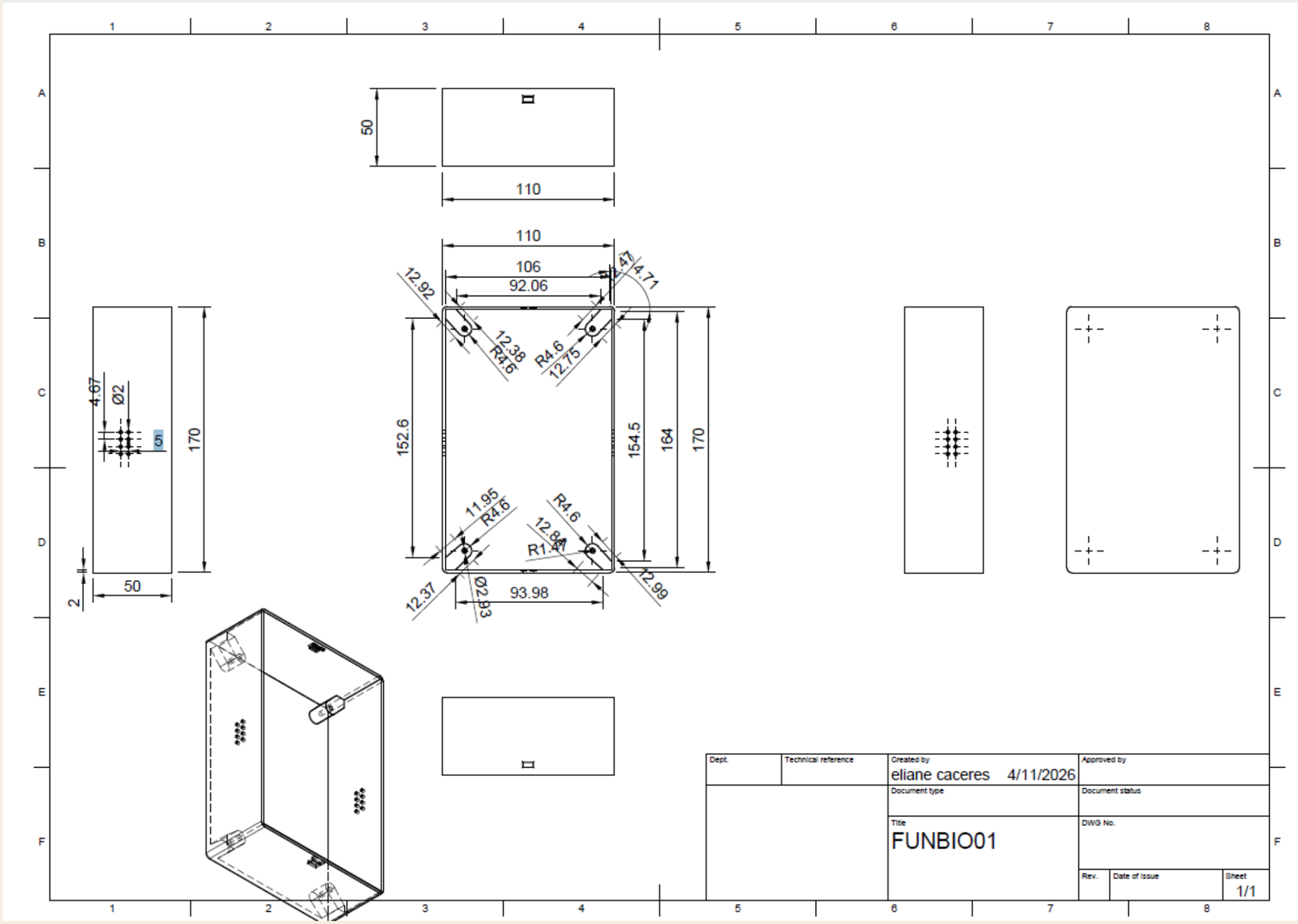






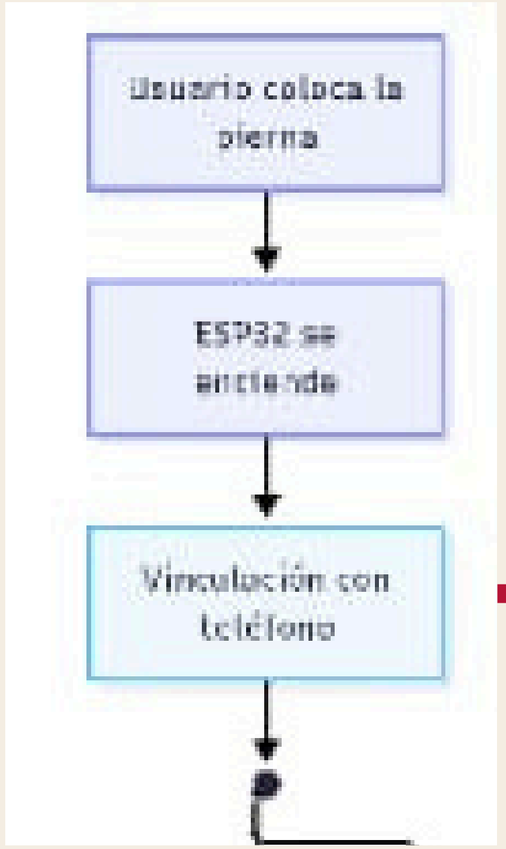
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CAJA ELECTRÓNICA



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PROGRAMACION Y CONTROL




```
NimBLEDevice::init("ESP32_BIOMEDICO");

bleServer = NimBLEDevice::createServer();

// Volver a anunciarse tras desconexión
bleServer->advertiseOnDisconnect(true);

NimBLEService* servicio =
    bleServer->createService(SERVICE_UUID);

chrTelemetria = servicio->createCharacteristic(
  TELEMETRY_UUID,
  NIMBLE_PROPERTY::READ | NIMBLE_PROPERTY::NOTIFY
);

NimBLECharacteristic* chrComando =
    servicio->createCharacteristic(
      COMMAND_UUID,
      NIMBLE_PROPERTY::WRITE
    );

chrEstado = servicio->createCharacteristic(
  STATUS_UUID,
  NIMBLE_PROPERTY::READ | NIMBLE_PROPERTY::NOTIFY
);
```

```
  servicio->start();

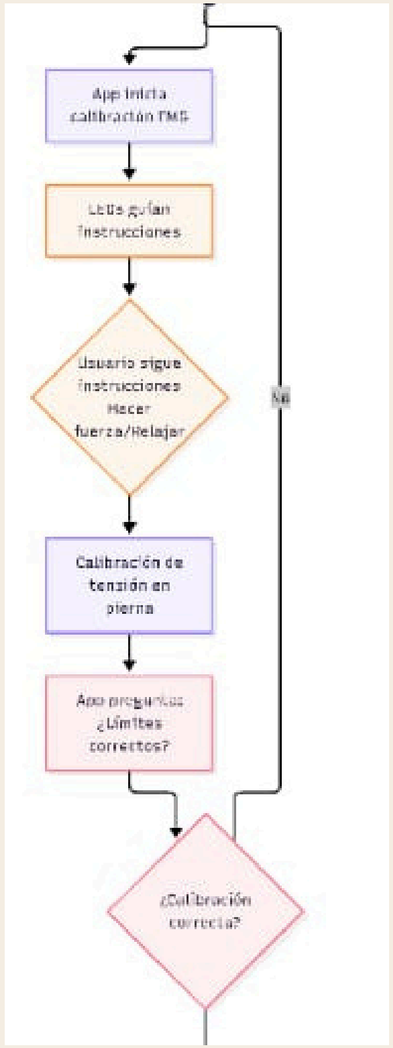
  NimBLEAdvertising* publicidad =
      NimBLEDevice::getAdvertising();

  publicidad->addServiceUUID(SERVICE_UUID);
  publicidad->start();

  Serial.println("ESP32 BLE listo");
}
```

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PROGRAMACION Y CONTROL




```
void ejecutarComando(const char* comando) {

  if (strcmp(comando, "EMG_START") == 0) {
    emgCalibraOrd = true;
    ultimaCalemg = micros();
    publicarEstado("Calibra_Inicio");
  }
```

```
void CalibrarEMG() {
  if (!emgCalibraOrd) return;
  delay(3000);
  emgMin = obtenerPromedio(TIEMPO_CALIBRACION);
  delay(3000);
  emgMax = obtenerMaximo(TIEMPO_CALIBRACION);
  emgActivo = true;
  publicarEstado("Calibra_Termino");
  emgCalibraOrd = false;
}
```

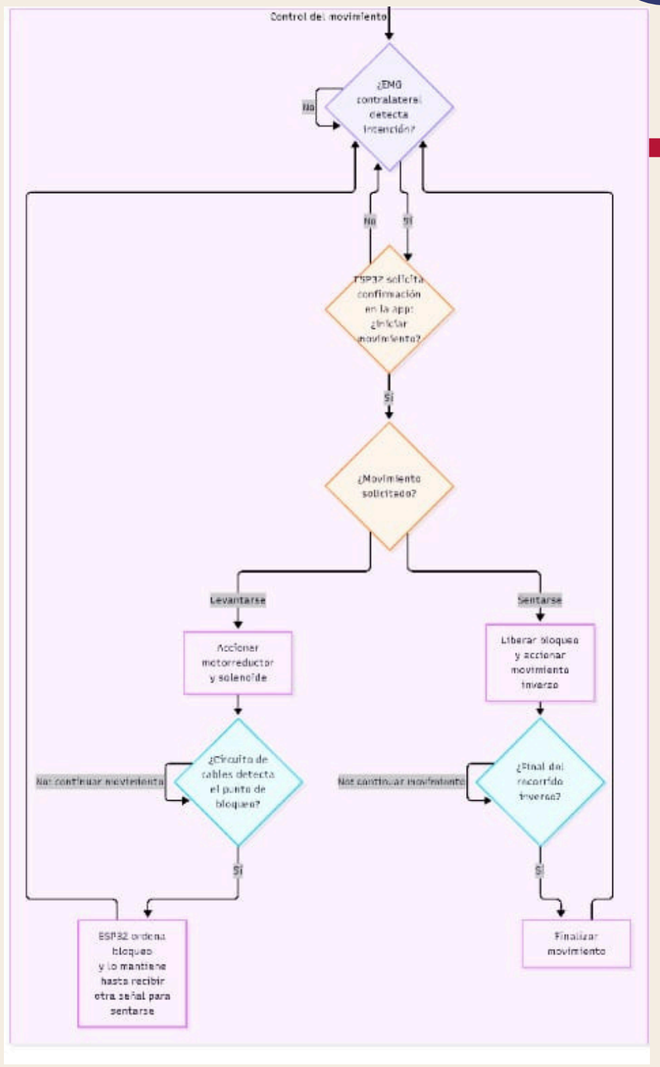
```
int obtenerMaximo(unsigned long duracion) {
  unsigned long inicio = millis();
  int maximo = 0;
  while (millis() - inicio < duracion) {
    int lectura = analogRead(PIN_EMG);
    if (lectura > maximo) {
      maximo = lectura;
    }
    delay(INTERVALO_MUESTREO);
  }

  return maximo;
}
```

```
int obtenerPromedio(unsigned long duracion) {
  unsigned long inicio = millis();
  uint32_t suma = 0;
  uint32_t muestras = 0;
  while (millis() - inicio < duracion) {
    int lectura = analogRead(PIN_EMG);
    suma += lectura;
    muestras++;
    delay(INTERVALO_MUESTREO);
  }
  return suma / muestras;
}
```

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PROGRAMACION Y CONTROL



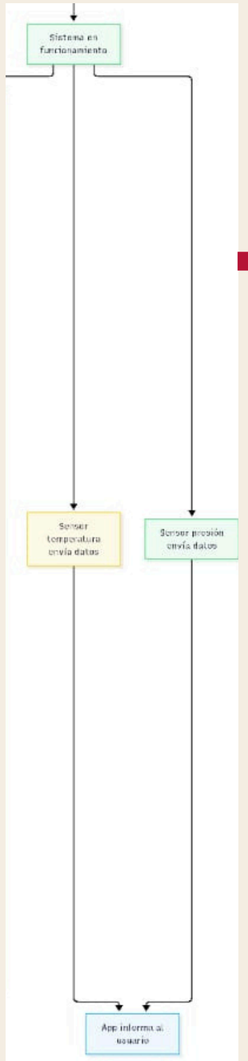

```
if (emgActivo) {
  int lecturaEMG = analogRead(PIN_EMG);
  AccionarApoyo(lecturaEMG);
}
```

```
void confirmacion (unsigned long inicio) {
  unsigned long actual;
  if(flexion && digitalRead(PIN_IFC) == HIGH) {
    do {
      actual = millis();
    } while (digitalRead(PIN_IFE) != HIGH || (actual-inicio) < 5000);
    analogWrite(PIN_HRV, 0);
    analogWrite(PIN_HLV, 0);
  } else if (!flexion && digitalRead(PIN_IFE) == HIGH) {
    do {
      actual = millis();
    } while (digitalRead(PIN_IFC) != HIGH || (actual-inicio) < 5000);
    analogWrite(PIN_HLV, 0);
    analogWrite(PIN_HRV, 0);
  }
}
```

```
void AccionarApoyo(int Lectura) {
    unsigned long inicio;
  if (!emgActivo) return;
  int porcentaje = map(
    Lectura,
    emgMin,
    emgMax,
    0,
    100
  );
  porcentaje = constrain(porcentaje, 0, 100);
  if (porcentaje > 75){
    if(flexion) {
      analogWrite(PIN_HRV, 254);
      inicio = millis();
      confirmacion(inicio);
      flexion = false;
    } else {
      analogWrite(PIN_HLV, 254);
      inicio = millis();
      confirmacion(inicio);
      flexion = true;
    }
  }
}
```

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PROGRAMACION Y CONTROL




```
uint32_t ahora = millis();

if (bleServer->getConnectedCount() > 0 &&
    (uint32_t)(ahora - ultimoEnvio) >= intervaloBLE) {

  ultimoEnvio = ahora;
  transmitirSensores();
}
```

```
//Lectura Temperatura
float leerTemperaturaC(int Pin) {
  unsigned long Tiempo_Muestreo = 10;
  return (obtenerPromedioSensor(Tiempo_Muestreo, PIN_LM[Pin])/10);
}

//Lectura FSR
uint16_t leerPresionADC(int Pin) {
  unsigned long Tiempo_Muestreo = 10;
  return obtenerPromedioSensor(Tiempo_Muestreo, PIN_FSR[Pin]);
}
```

```
void transmitirSensores() {
  int i = 0;
  while (i<5) {
    if (i<2) {
      float temperatura[i] = leerTemperaturaC(PIN_LM[i]);
      uint16_t presionADC[i] = leerPresionADC(PIN_FSR[i]);
    } else {
      uint16_t presionADC[i] = leerPresionADC(PIN_FSR[i]);
    }
    i++;
  }

  char mensaje[64];

  snprintf(
    mensaje, sizeof(mensaje),
    "T,%.1f,%.1f,P,%u,%u,%u,%u,%u",
    temperatura[0], temperatura[1], presionADC[0], presionADC[1], presionADC[2], presionADC[3], presionADC[4]
  );

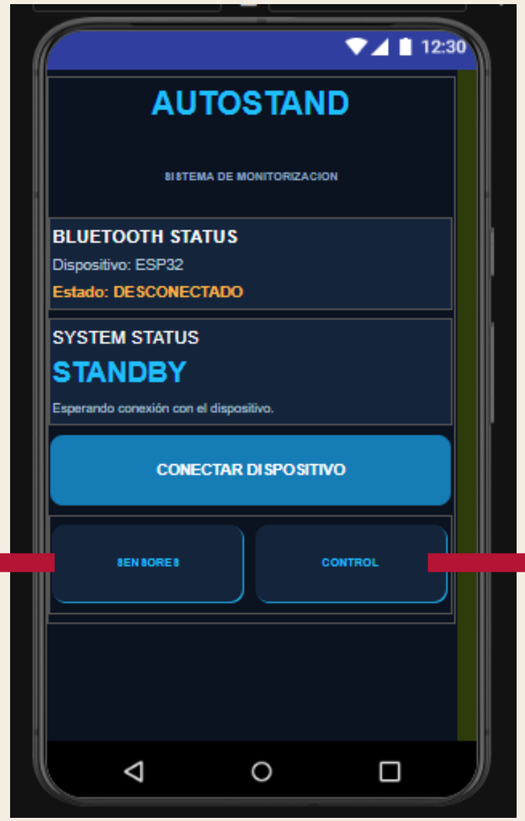
  chrTelemetria->setValue(mensaje);
  chrTelemetria->notify();
}
```

```
//Filtrado de lecturas
int obtenerPromedioSensor(unsigned long duracion, int PIN) {
  unsigned long inicio = millis();
  uint32_t suma = 0;
  uint32_t muestras = 0;
  while (millis() - inicio < duracion) {
    int lectura = analogReadMilliVolts(PIN);
    suma += lectura;
    muestras++;
  }
  return suma / muestras;
}
```

INTERFACES DIGITALES

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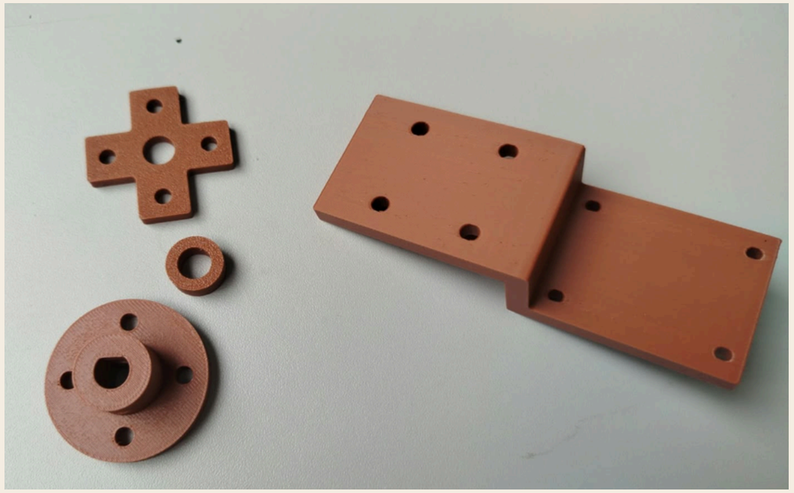


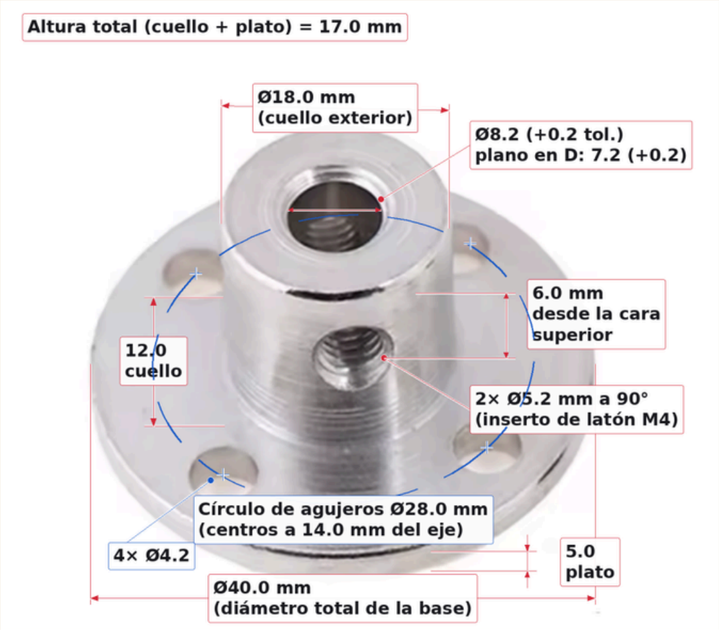


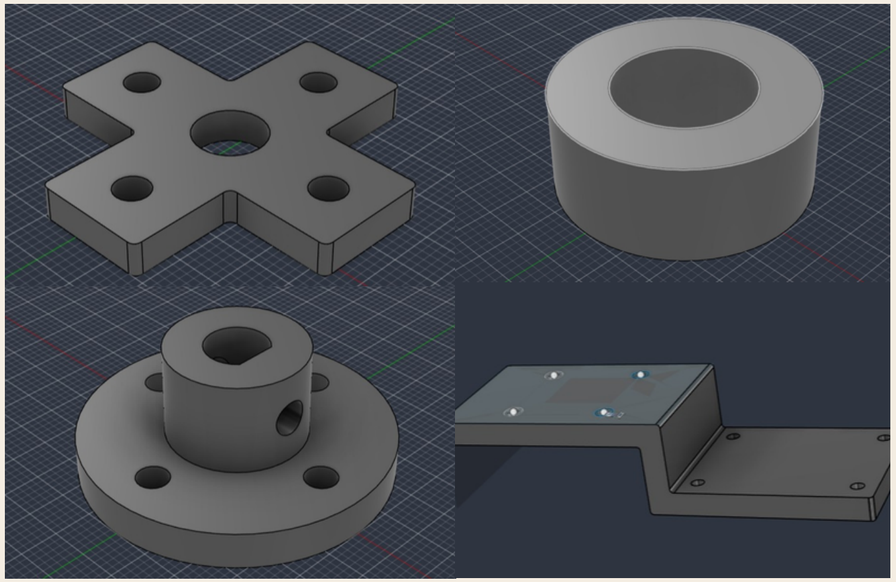


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Mecánica y fabricación

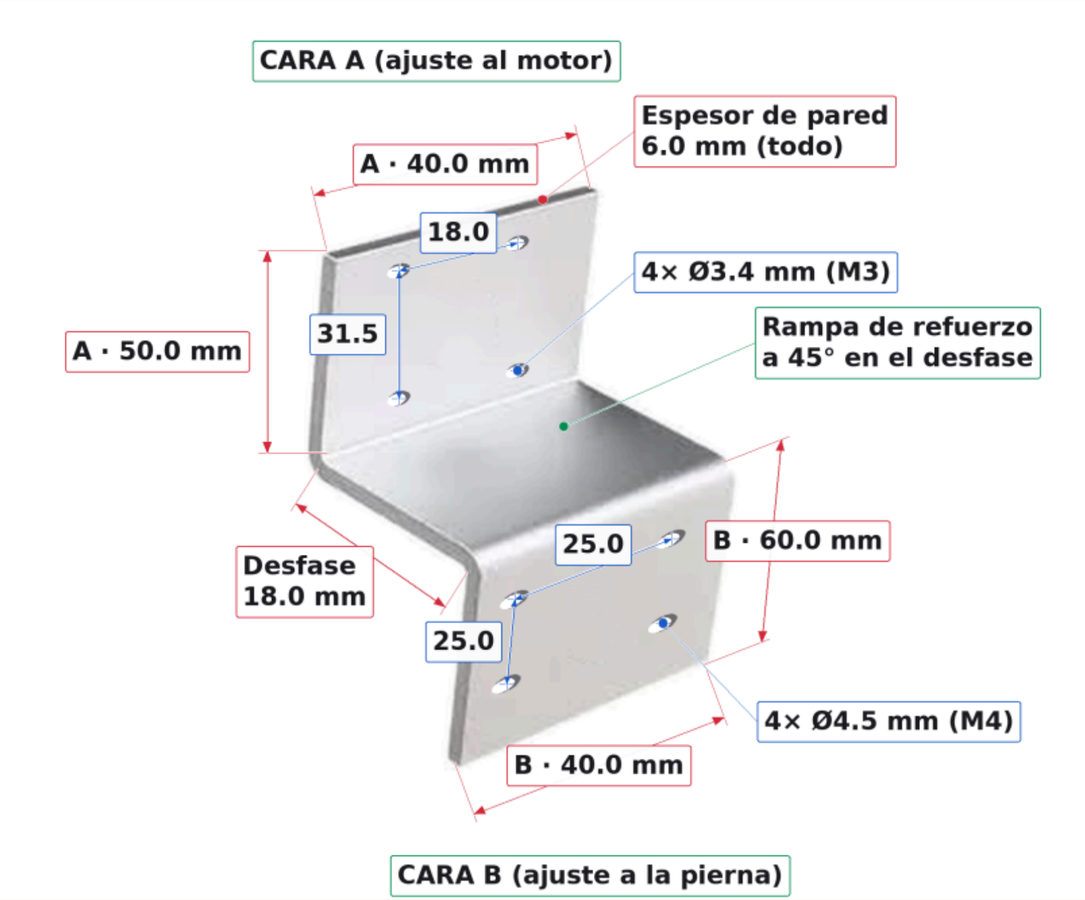


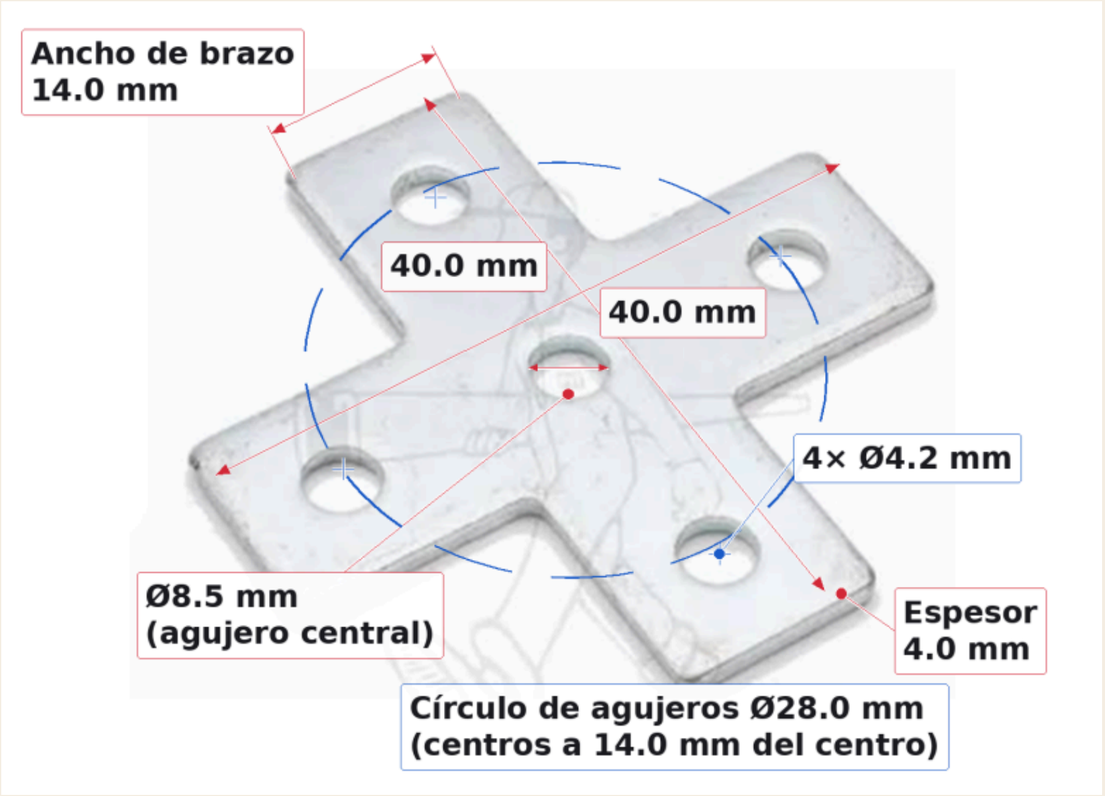






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GRACIAS